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Tut 6.

2) Exercise 1

1) Convolution with stride 1,

Filter : $K = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 1 \end{bmatrix}$

$$I = \begin{bmatrix} 20 & 35 & 35 & 35 & 20 \\ 29 & 46 & 44 & 42 & 27 \\ 16 & 25 & 21 & 19 & 12 \\ 66 & 120 & 116 & 114 & 62 \\ 74 & 216 & 174 & 172 & 112 \\ 70 & 210 & 170 & 170 & 110 \end{bmatrix}$$

$C(1,1) = 20+35+35+29+44+16+25+21 = 225$

$C(1,2) = 35+35+35+46+42+25+21+19 = 258$

$C(1,3) = 250$, $C(1,4) = 209$, $C(2,1) = 483$, $C(2,2) = 566$.

$\therefore$ Convolution Output

$$O = \begin{bmatrix} 255 & 258 & 250 & 209 \\ 458 & 566 & 552 & 472 \\ 708 & 982 & 1320 & 802 \\ 1000 & 1488 & 1320 & 1224 \end{bmatrix}$$

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2) Max pooling

$$I = \begin{bmatrix} 225 & 258 & 250 & 209 \\ 458 & 566 & 552 & 472 \\ 708 & 981 & 1320 & 802 \\ 1000 & 1488 & 1320 & 1224 \end{bmatrix}$$

~~C(1,1) =~~

C(1,1) = 566
C(1,2) = 552
C(2,1) = 1488
C(2,2) = 1320.

$$\therefore \text{Output} = \begin{bmatrix} 566 & 562 \\ 1488 & 1320 \end{bmatrix}$$

3) Max pooling on 4x4 feature map.

$$I = \begin{bmatrix} 5 & 12 & 8 & 1 \\ 3 & 7 & 6 & 14 \\ 9 & 4 & 2 & 11 \\ 15 & 10 & 13 & 16 \end{bmatrix}$$

C(1,1) = 12
C(1,2) = 14
C(2,1) = 15
C(2,2) = 16.

$$\therefore \text{Output} = \begin{bmatrix} 12 & 14 \\ 15 & 16 \end{bmatrix}$$

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Q.4 Average pooling

$$X = \begin{bmatrix} 1 & 2 & 3 & 4 \\ 5 & 6 & 7 & 8 \\ 9 & 10 & 11 & 12 \\ 13 & 14 & 15 & 16 \end{bmatrix}$$

$c(1,1) = 1+2+5+6/4 = 3.5$
$c(1,2) = 3+4+7+8/4 = 5.5$
$c(2,1) = 11.5$
$c(2,2) = 13.5$

$\therefore$ Output $= \begin{bmatrix} 3.5 & 5.5 \\ 11.5 & 13.5 \end{bmatrix}$

Exercise 2: Parameter count.

(a) Conv2D layer 1:
- weights : $3 \times 3 \times 3 \times 16 = 432$
- bias = 16.

$\therefore$ Total $= 432 + 16 = 448$.

b) Batch Normalization layer : 2. (16 Feature maps)
$\therefore$ Parameters $= 16 \times 2 = 32$.

c) Maxpooling.
- no parameters.

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d) Conv2D layer 2:
- weights = $3 \times 3 \times 16 \times 32 = 4608$
- bias = 32

Total = $4608 + 32 = 4640$

e) Batch Normalization layer 2.
Total = $32 \times 2 = 64$.

f) Max pooling
- No parameters

g) Flattening - $8 \times 8 \times 32 = 2048$
- No parameters

h) Dense layer 1 : Input = 2048 Output = 128
Weight = $2048 \times 128 = 262144$
Bias = 128
Total = 262272

i) Dense layer 2.
Input = 128 Output 10
Weight = $128 \times 10 = 1280$
Bias = 10.
Total = $1280 + 10 = 1290$